

Name: _____

Solutions

Date: _____



Year 9 Mathematics

Test 6, 2015

Topics – Scientific notation, congruency and similarity, algebra.

<u>51</u> = _____ %

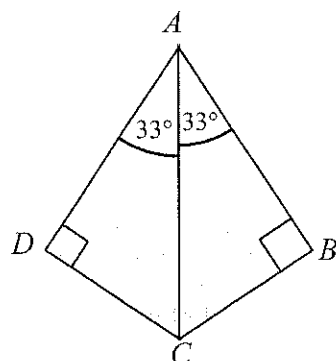
Total Time: **55 minutes**
 Weighting: **9% of the year.**
 Equipment Allowed: **page of notes (A4 one side), calculator**

CALCULATOR ALLOWED

Part A – Multiple Choice [4 marks: 1 each]

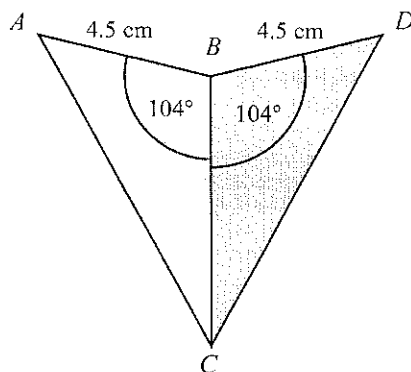
For the following multiple-choice questions, circle the correct answer.

1. Which of the congruence tests could be used to show that $\triangle ADC \equiv \triangle ABC$?



- ☒ AAS ☐ RHS ☐ SAS ☐ SSS

2. Which of the congruence tests could be used to show that $\triangle ABC \equiv \triangle DBC$?



- ☐ AAS ☐ RHS ☒ SAS ☐ SSS

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3. An equilateral triangle of side length 15 cm is reduced to become a triangle of side length 5 cm. The scale factor used is:

☒ $\frac{1}{3}$ ✓
 ☐ 5
 ☐ $\frac{1}{5}$
 ☐ 115

4. Robin takes a photograph, which measures 8 cm by 10 cm, to the printers to make a poster. The poster is 8 times the size of the photograph. What is the measurement of the poster on its shorter side?

☒ 64 cm ✓
 ☐ 72 cm
 ☐ 80 cm
 ☐ 160 cm

Part B – Short-answer questions

5. [3 marks: 1, 1, 1]

Use words from the list below to complete the following sentences.

scale factor
 similar
 congruent
 figure
 side

- a) Two triangles are congruent ✓ if they are the same shape and the same size.
 b) Triangles that are the same shape but not the same size are similar ✓.
 c) The scale factor ✓ is the ratio between the sides of two similar figures.

6. [5 marks: 1, 1, 1, 1, 1]

- a) Write the following numbers in scientific notation:

i) 92 017 000

$$9.2017 \times 10^7$$

$\frac{1}{2}$ $\frac{1}{2}$

ii) 0.000 056 2

$$5.62 \times 10^{-5}$$

$\frac{1}{2}$ $\frac{1}{2}$

- b) Write the value of the following numbers written in scientific notation

i) 3.2×10^4

$$32000$$
 ✓

ii) 7.23×10^{-3}

$$0.00723$$
 ✓

- c) Evaluate $4.56 \times 10^{-3} \div 5.89 \times 10^{-8}$. Express your answer in standard form.

$$7.741935484 \times 10^4$$

$\frac{1}{2}$ $\frac{1}{2}$

7. [3 marks: 1, 1, 1]

a) How many significant figures does the number 23.40 have?

4 ✓

b) How many significant figures does the number 0.0042 have?

2 ✓

c) Round 13.9998 to four significant figures.

14.00 ✓

8. [2 marks, ½ each]

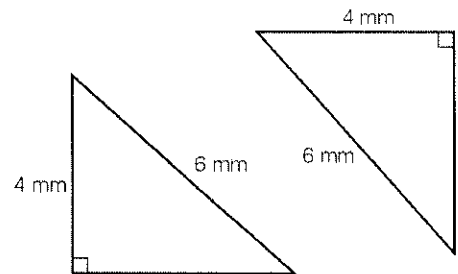
Give reasons why the following pair of triangles are congruent.

The hypotenuse in each triangle = 6 mm ✓

Each triangle has another side equal to 4 mm ✓

Each triangle has a Right angle ✓

The triangles are congruent using the RHS test. ✓



9. [4 marks: 1, 1, 2]

Simplify the following

a. $3s + 4s$

$7s$ ✓

b. $8f - 5f + 2f$

$5f$ ✓

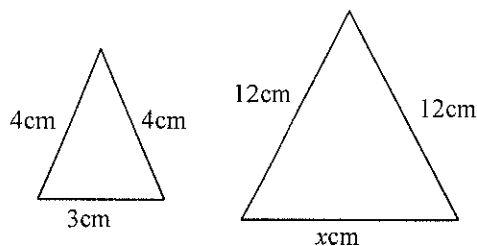
$2d + 5g + 12d - 4g$

$14d - 9g$ ✓

10. [4 marks: 2, 2]

Determine the value of x in each of the following pairs of similar triangles, and state the rule used (AAA, SSS, SAS, RHS).

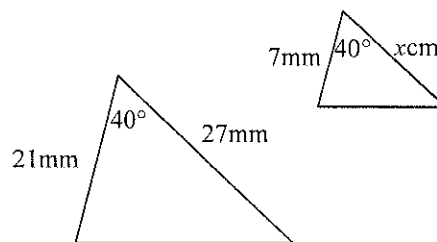
a.



Rule: SSS ✓

$x = 9\text{cm}$ ✓

b.



Rule: SAS ✓

$x = 9\text{cm}$ ✓

11. [11 marks: 2, 2, 2, 2, 3]

Expand and simplify the following

a. $4(d+3)$

$$4d + 12$$

b. $-2(e+5)$

$$-2e - 10$$

c. $k(k-4)$

$$k^2 - 4k$$

d. $-3h(h-7)$

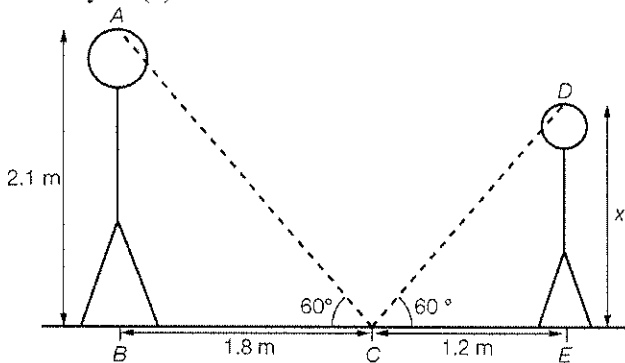
$$-3h^2 + 21h$$

e. $(r+3)(r+2)$

$$r^2 + 5r + 6$$

12. [5 marks]

Christy's dad bounces a ball from his head height and Christy catches it at her head height, as the following diagram shows. Prove the following triangles are similar and find how tall Christy is (x).



Statement

Reason

$$\angle BCA = \angle ECD$$

Given ✓

$$\angle ABC = \angle DEC$$

Right angles ✓

$$\angle BAC = \angle EDC$$

Angles in a triangle ✓

$$\therefore \triangle ABC \sim \triangle CDE$$

AAA ✓

$$\text{Scale factor} = \frac{1.2}{1.8} = \frac{2}{3} \checkmark$$

$$x = \frac{2}{3} \times 2.1$$

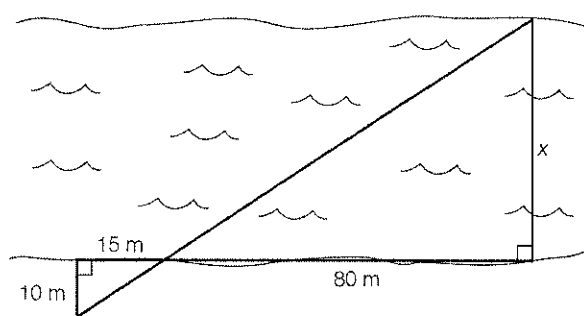
$$= 1.4 \text{ m}$$

Christy is 1.4 m tall ✓

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13. [2 marks]

Use the properties of similar triangles to find the distance across the river, correct to one decimal place.



$$\text{Scale factor} = \frac{80}{15} = \frac{16}{3} = 5\frac{1}{3} \checkmark$$

$$x = 10 \times 5\frac{1}{3} = 53.3 \text{ m} \checkmark$$

14. [8 marks: 2, 2, 2, 2]

Factorise the following

a. $3d + 12$

$$3(d+4) \checkmark \checkmark$$

b. $p^2 - 7p$

$$p(p-7) \checkmark \checkmark$$

c. $w^2 + 15w + 50$

$$(w+10)(w+5) \checkmark \checkmark$$

e. $h^2 - 9$

$$(h+3)(h-3) \checkmark \checkmark$$

~ END OF TEST ~

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